

2

Answer: B

Let x be the distance travelled by A and B.

$$\text{Ball A: } x = vt - 0.5g\sin\theta t^2 \text{ --- (1)}$$

$$\text{Ball B: } x = 0.5g\sin\theta t^2 \text{ --- (2)}$$

Solving (1) and (2),

$$v = g(\sin 30^\circ)(2) = 9.81 \text{ m s}^{-1}$$

2

Answer: B